Higher Top-Up Prelim Revision by Topic

The Higher Webpage contains hints and worked solutions for these questions, as well as **practice papers** to help you prepare for the March Top-Up Prelim.

FORMULAE LIST

Circle

$x^2 + y^2 + 2gx + 2fy + c = 0$ represents a circle centre $(-g, -f)$, radius $\sqrt{g^2 + f^2 - c}$.

$(x - a)^2 + (y - b)^2 = r^2$ represents a circle centre (a, b) and radius r.

Scalar product

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta, \text{ where } \theta \text{ is the angle between } \mathbf{a} \text{ and } \mathbf{b}$$

or

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 \text{ where } \mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

Trigonometric formulae

$$\begin{aligned} \sin(A \pm B) &= \sin A \cos B \pm \cos A \sin B \\ \cos(A \pm B) &= \cos A \cos B \mp \sin A \sin B \\ \sin 2A &= 2 \sin A \cos A \\ \cos 2A &= \cos^2 A - \sin^2 A \\ &= 2\cos^2 A - 1 \\ &= 1 - 2\sin^2 A \end{aligned}$$

Table of standard derivatives

$f(x)$	$f'(x)$
$\sin ax$	$a \cos ax$
$\cos ax$	$-a \sin ax$

Table of standard integrals

$f(x)$	$\int f(x)\, \mathrm{d}x$
$\sin ax$	$-\frac{1}{a} \cos ax + C$
$\cos ax$	$\frac{1}{a} \sin ax + C$

Top-Up Prelim Topic List

13. The Circle (Continued)

14. Wave Function

15. Logs and Exponentials

16. Further Calculus

⋆ Answers

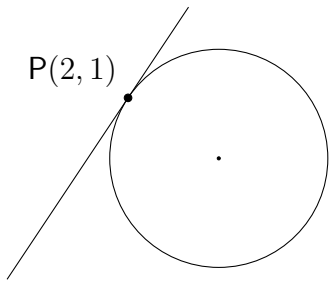
Note: Questions which require a **calculator** are marked with 🖩.

13 The Circle (Continued)

75. Find the coordinates of the points of intersection of the circle with equation $x^2 + y^2 - 8x + 4y - 5 = 0$ and the line with equation $x + y = 3$.

76. Show that the line with equation $y = x - 3$ is a tangent to the circle with equation $x^2 + y^2 - 6x + 4y + 11 = 0$ and find the coordinates of the point of contact.

77. Point P$(2, 1)$ lies on circle C with equation $x^2 + y^2 - 16x + 6y + 21 = 0$.



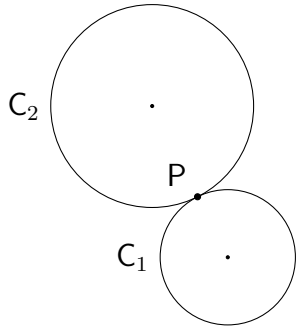
Find the equation of the tangent to the circle at P.

78. Circle C has equation $(x-5)^2+(y+3)^2=17$.

(a) State the radius and the centre of circle C.

(b) Show that the point Q$(9,-8)$ lies outside of circle C.

79. Circles C_1 and C_2 are shown in the diagram below.



Their equations are:

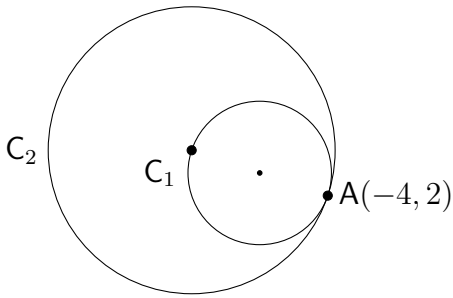
- $C_1: \quad (x-3)^2+(y+7)^2=20$
- $C_2: \quad x^2+y^2+4x-6y-32=0$

(a) Determine the distance between the centres of C_1 and C_2.

(b) Show that circles C_1 and C_2 touch externally.

(c) The point of contact, P, is shown on the diagram.
Determine the coordinates of P.

80. Circle C_1 touches circle C_2 internally at point A$(-4,2)$ and passes through the centre of circle C_2, as shown in the diagram below.



Circle C_1 has equation $x^2+y^2+14x-6y+48=0$.
Determine the equation of circle C_2.

14 Wave Function

For all questions, $k > 0$ and 🖩.

81. Express $8\sin x^\circ + 7\cos x^\circ$ in the form $k\sin(x+a)^\circ$, $0 < a < 360$.

82. Express $\sqrt{5}\sin x + \cos x$ in the form $k\cos(x-a)$, $0 < a < 2\pi$.

83. (a) Express $2\sin x^\circ - 4\cos x^\circ$ in the form $k\sin(x-a)^\circ$, $0 < x < 360$.

    (b) Hence state:

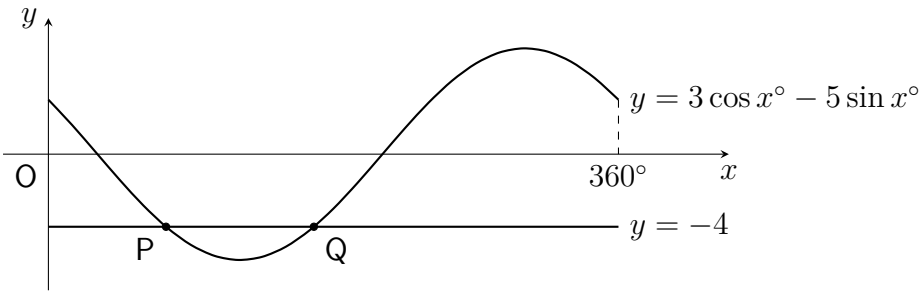
        i. The maximum value of $2\sin x^\circ - 4\cos x^\circ$.

        ii. The value of x for which it occurs, where $0 < x < 360$.

84. (a) Express $2\cos x^\circ - \sin x^\circ$ in the form $k\cos(x+a)^\circ$, $0 < a < 360$.

    (b) Hence solve $2\cos x^\circ - \sin x^\circ = -1$ where $0 < x < 360$.

85. Part of the graphs of $y = 3\cos x^\circ - 5\sin x^\circ$ and $y = -4$ are shown in the diagram below:



Points P and Q are points of intersection.

(a) Express $3\cos x^\circ - 5\sin x^\circ$ in the form $k\cos(x+a)^\circ$, $0 < a < 360$.

(b) Hence determine the coordinates of P and Q.

15 Logs and Exponentials

86. Express $2\log_p 10 + \log_p 2 - \log_p 50$ in the form $\log_p q$.

87. Evaluate the following:

    (a) $\log_3 \frac{1}{4} + 2\log_3 6$

    (b) $\frac{1}{2}\log_5 64 - \log_5 40$

88. Solve each equation:

    (a) $\log_6 x + \log_6 4 = 2$

    (b) $\log_2 x + \log_2 (x+7) = 3$

89. Given that $\log_p 4 + 3 = \log_p 32$, find the value of p.

90. A population of mice is discovered living in some woodland. The number of mice can be modelled by:

$$M = 80e^{kt}$$

    where M is the estimated number of mice in the population, t is the number of weeks since the population was discovered and k is a constant.

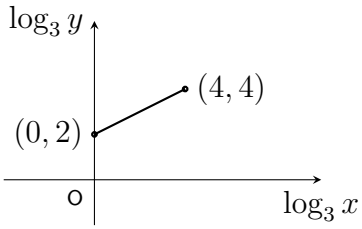
    (a) State the estimated number of mice in the population when it was first discovered.

    (b) It is estimated there will be 120 mice in the population after 3 weeks. Find the value of k.

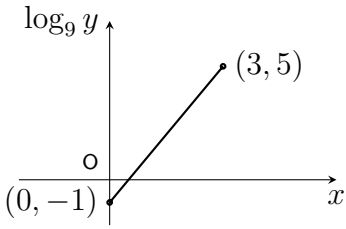
    (c) Calculate the time taken for the population to double in size.

91. Variables x and y are linked by the equations given below. Find the values of a and b for each:

    (a) $y = ax^b$



    (b) $y = ab^x$



16 Further Calculus

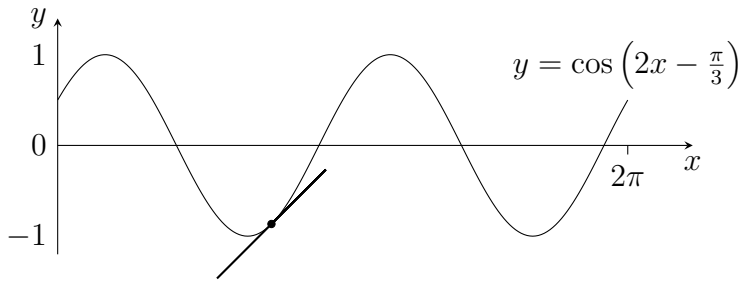
92. Given that $y = (6x-5)^3$, find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$.

93. A function, f, is defined on a suitable domain by $f(x) = \dfrac{1}{(3x-5)^2}$. Determine the rate of change of f when $x = 2$.

94. Given $g(x) = 4\cos 3x + 2\sin 2x$, find the value of $g'\left(\frac{\pi}{2}\right)$.

95. Differentiate $\sqrt[3]{6x^2-1}$ with respect to x.

96. Find the gradient of the tangent to $y = \cos\left(2x - \frac{\pi}{3}\right)$ when $x = \frac{3\pi}{4}$.



97. Find $\displaystyle\int (2x-3)^5 \, \mathrm{d}x$.

98. Evaluate $\displaystyle\int_0^1 2\cos 3x \, \mathrm{d}x$.

99. Find $\displaystyle\int \frac{3}{(x+4)^{\frac{1}{2}}} \, \mathrm{d}x$.

100. Find the value of $\displaystyle\int_0^{\frac{\pi}{3}} \sin\left(x + \frac{\pi}{6}\right) \, \mathrm{d}x$.

Answers

Answers 13 The Circle

75. $(1, 2)$ and $(8, -5)$

76. $(2, -1)$ and statement (*"Show that..."*)

77. $y = \frac{3}{2}x - 2$ or $2y = 3x - 4$

78. (a) Centre $(5, -3)$ and radius $\sqrt{17}$

    (b) $\sqrt{41} > \sqrt{17}$ and conclusion, or other valid approach

79. (a) $\sqrt{125}$ or $5\sqrt{5}$

    (b) $3\sqrt{5} + 2\sqrt{5} = 5\sqrt{5}$ and conclusion, or other valid approach

    (c) $(1, -3)$

80. $(x + 10)^2 + (y - 4)^2 = 40$

Answers 14 Wave Function

81. $\sqrt{113}\sin(x + 41.2)^\circ$

82. $\sqrt{6}\cos(x - 1.15)$

83. (a) $2\sqrt{5}\sin(x - 63.4)^\circ$

    (b) i. max value is $2\sqrt{5}$

        ii. occurs when $x = 153.4$

84. (a) $\sqrt{5}\cos(x + 26.6)^\circ$

    (b) $x = 90, 216.8$

85. (a) $\sqrt{34}\cos(x + 59.0)^\circ$

    (b) P$(73.3^\circ, -4)$ and Q$(167.7^\circ, -4)$

Answers 15 Logs and Exponentials

86. $\log_p 4$

87. (a) 2

    (b) -1

88. (a) $x = 9$

    (b) $x = 1$

89. $p = 2$

90. (a) 80

    (b) $k = 0.135$

    (c) 5.13 weeks

91. (a) $a = 9$
    $b = \frac{1}{2}$

    (b) $a = \frac{1}{9}$
    $b = 81$

Answers 16 Further Calculus

92. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 18(6x-5)^2$

93. -6

94. 8

95. $4x(6x^2-1)^{-\frac{2}{3}}$

96. 1

97. $\frac{(2x-3)^6}{12} + \mathrm{C}$

98. 0.0941

99. $6(x+4)^{\frac{1}{2}} + \mathrm{C}$

100. $\frac{\sqrt{3}}{2}$